Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:

- Check your working directory
- Load the tidyverse, lubridate, zoo, and trend packages
- Set your ggplot theme

```
library(here)
```

```
## here() starts at /home/guest/ENVIRON 872/EDE_Fall2025
```

```
here()
```

```
## [1] "/home/guest/ENVIRON 872/EDE_Fall2025"
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.1.0
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(lubridate)
library(zoo)
```

```
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
library(trend)

#my theme
mytheme_08<- theme_minimal() +
  theme(axis.text = element_text(color = "black"),
        legend.position = "right")
theme_set(mytheme_08)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named `GaringerOzone` of 3589 observation and 20 variables.

```r
#1
df1 <-read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2010_raw.csv"))
df2 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2011_raw.csv"))
df3 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2012_raw.csv"))
df4 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2013_raw.csv"))
df5 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2014_raw.csv"))
df6 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2015_raw.csv"))
df7 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2016_raw.csv"))
df8 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2017_raw.csv"))
df9 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2018_raw.csv"))
df10 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2019_raw.csv"))

GaringerOzone <- bind_rows(df1, df2, df3, df4, df5, df6, df7, df8, df9, df10)
```

Wrangle

3. Set your date column as a date class.

4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.

5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".

6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```
# 3
GaringerOzone$Date <- as.Date(GaringerOzone$Date, format = '%m/%d/%Y')

class(GaringerOzone$Date)
```

```
## [1] "Date"
```

```
# 4
GaringerOzone_wrangled <- GaringerOzone %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)

# 5
Days <- as.data.frame(seq(from = as.Date('2010-01-01'),
                          to = as.Date('2019-12-31'),
                          by = 'day'))
names(Days) <- 'Date'

# 6
GaringerOzone <- left_join(Days, GaringerOzone_wrangled,
                           by = "Date")
summary(GaringerOzone)
```

```
##       Date            Daily.Max.8.hour.Ozone.Concentration DAILY_AQI_VALUE
##  Min.   :2010-01-01   Min.   :0.00200                      Min.   :  2.00
##  1st Qu.:2012-07-01   1st Qu.:0.03200                      1st Qu.: 30.00
##  Median :2014-12-31   Median :0.04100                      Median : 38.00
##  Mean   :2014-12-31   Mean   :0.04163                      Mean   : 41.57
##  3rd Qu.:2017-07-01   3rd Qu.:0.05100                      3rd Qu.: 47.00
##  Max.   :2019-12-31   Max.   :0.09300                      Max.   :169.00
##                       NA's   :63                           NA's   :63
```

Visualize

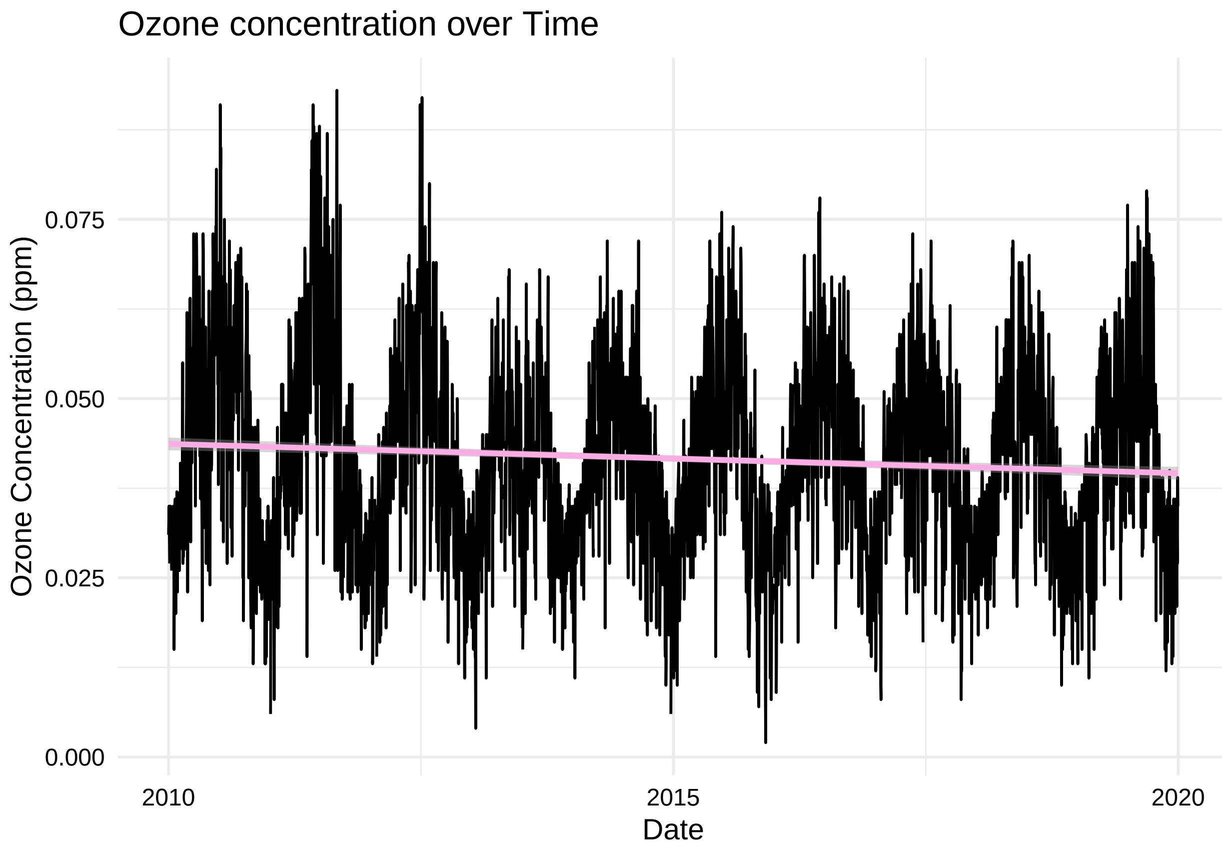
7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7

GaringerOzone %>%
  ggplot(aes(x = Date,
             y = Daily.Max.8.hour.Ozone.Concentration)) +
  geom_line() +
  labs(y = "Ozone Concentration (ppm)",
       title = "Ozone concentration over Time") +
  geom_smooth(method = 'lm',
              color = '#FCACE4') +
  mytheme_08
```

```
## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```



Answer: The smoothed line shows that there is a slight deecrease in ozone concentration from 2010 to 2019. The line is not steep indicating that this change is gradual and not rapid.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
GaringerOzone <-GaringerOzone %>%
  mutate(Daily.Max.8.hour.Ozone.Concentration = na.approx(Daily.Max.8.hour.Ozone.Concentration,
                                                          Date,
                                                          na.rm = FALSE))

summary(GaringerOzone)
```

```
##       Date               Daily.Max.8.hour.Ozone.Concentration DAILY_AQI_VALUE
##  Min.   :2010-01-01   Min.   :0.00200                      Min.   :  2.00
##  1st Qu.:2012-07-01   1st Qu.:0.03200                      1st Qu.: 30.00
##  Median :2014-12-31   Median :0.04100                      Median : 38.00
##  Mean   :2014-12-31   Mean   :0.04151                      Mean   : 41.57
##  3rd Qu.:2017-07-01   3rd Qu.:0.05100                      3rd Qu.: 47.00
##  Max.   :2019-12-31   Max.   :0.09300                      Max.   :169.00
##                                                            NA's   :63
```

Answer: We used a linear interpolation because ozone concentrations can change gradually over a day and linear interpolation draws a straight line between two measured values giving us a realistic estimate for the missing days without skewing from the observed pattern. We do not want to use piecewise since it uses the nearest neighbor measured value and that would make the missing ozone concentrations the same across days. Spline interpolation would also not be a good function to use because it uses quadratic functions, meaning a curved line through the data, which would not accurately represent the pattern in our measured data.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```r
#9
GaringerOzone.monthly <- GaringerOzone %>%
  mutate(year = year(Date),
         month = month(Date)) %>%
  group_by(year, month)

GaringerOzone.monthly <- GaringerOzone.monthly %>%
  mutate(Date = as.Date(paste0(year, '-', month, '-01'))) %>%
  group_by(Date) %>%
  summarize(Ozone.Concentration = mean(Daily.Max.8.hour.Ozone.Concentration))

summary(GaringerOzone.monthly)
```

```
##       Date               Ozone.Concentration
##  Min.   :2010-01-01   Min.   :0.02342
##  1st Qu.:2012-06-23   1st Qu.:0.03380
##  Median :2014-12-16   Median :0.04335
##  Mean   :2014-12-16   Mean   :0.04149
##  3rd Qu.:2017-06-08   3rd Qu.:0.04915
##  Max.   :2019-12-01   Max.   :0.06623
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

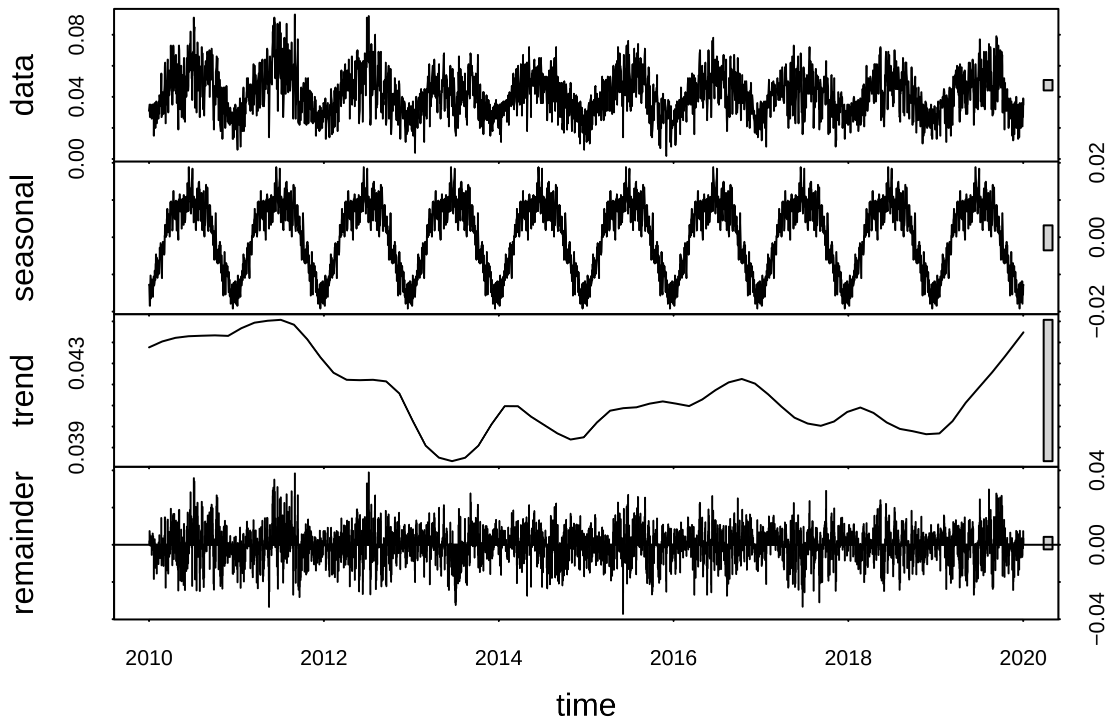
```r
#10
GaringerOzone.daily.ts <-ts(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration,
                            start = c(2010, 01, 01),
                            frequency = 365)
```

```r
GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly$Ozone.Concentration,
                               start = c(2010, 01),
                               frequency = 12)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

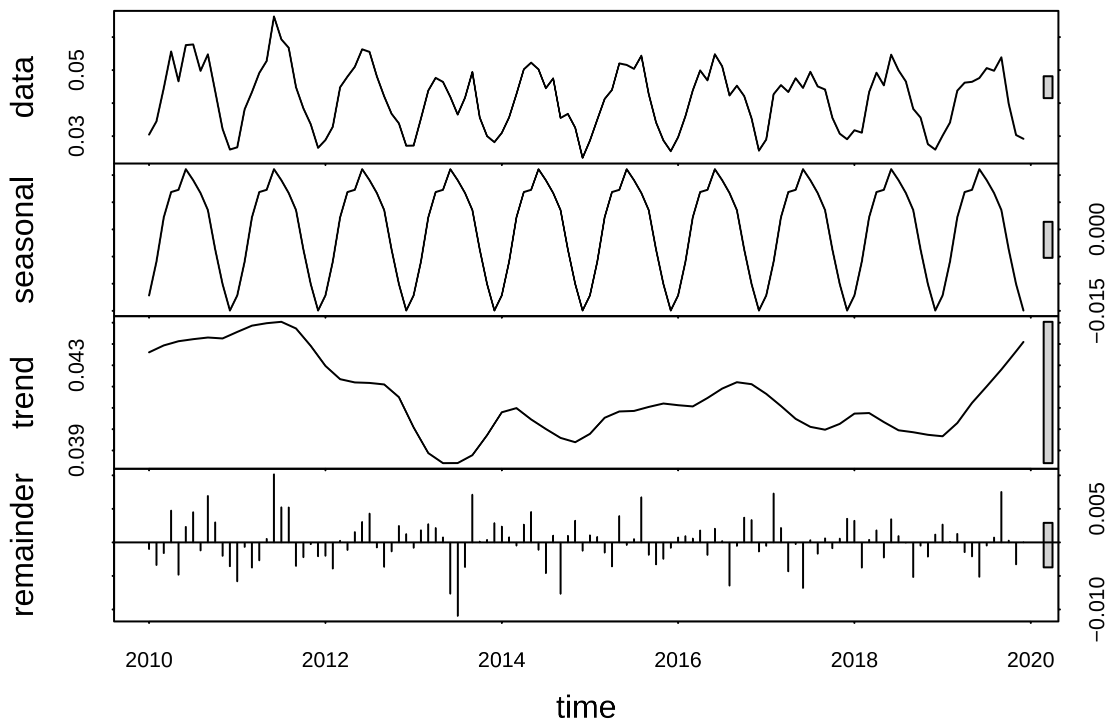
```r
#11
GaringerOzone.daily.decomposed <- stl(GaringerOzone.daily.ts,
                                      s.window = 'periodic')

plot(GaringerOzone.daily.decomposed)
```




```r
GaringerOzone.monthly.decomposed <- stl(GaringerOzone.monthly.ts,
                                        s.window = 'periodic')

plot(GaringerOzone.monthly.decomposed)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12
GaringerOzone.monthly.trend <- Kendall::SeasonalMannKendall(GaringerOzone.monthly.ts)

GaringerOzone.monthly.trend
```

```
## tau = -0.143, 2-sided pvalue =0.046724
```

```
summary(GaringerOzone.monthly.trend)
```

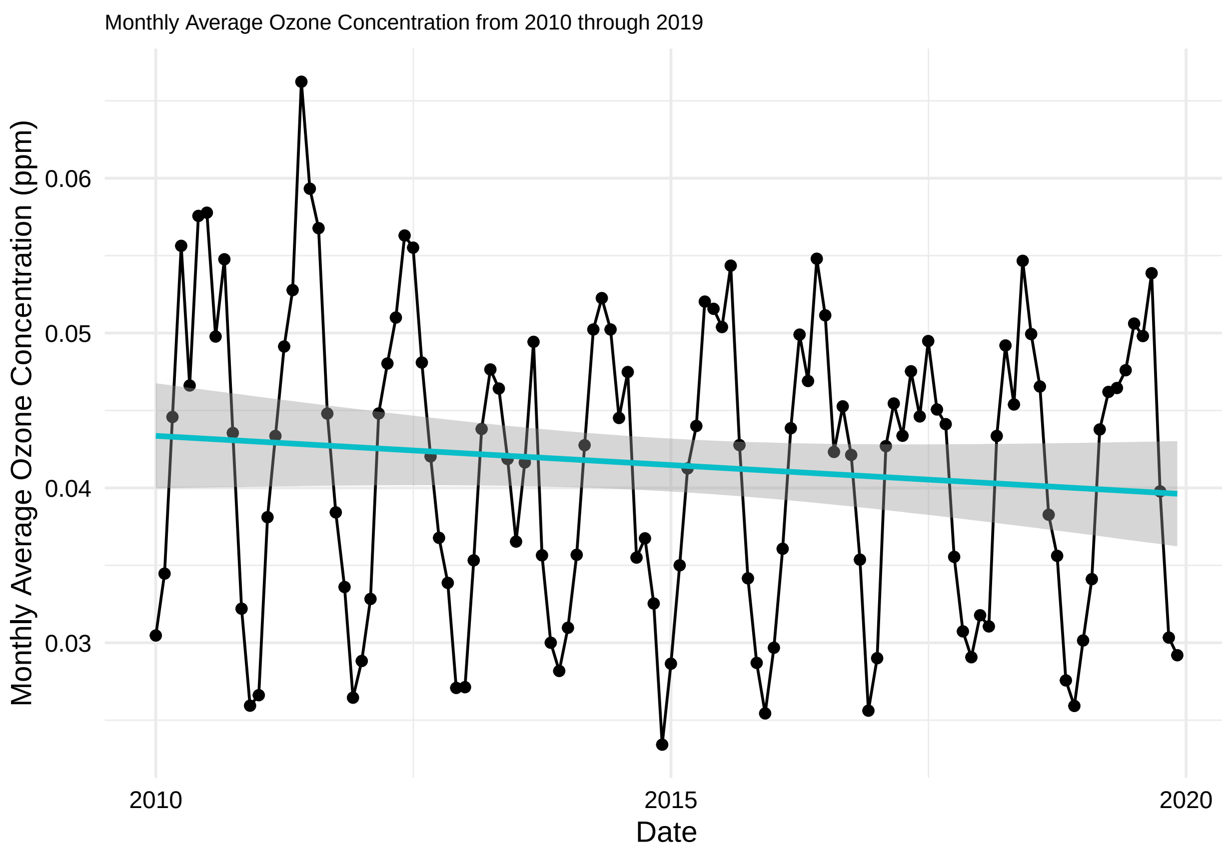
```
## Score =  -77 , Var(Score) = 1499
## denominator =  539.4972
## tau = -0.143, 2-sided pvalue =0.046724
```

Answer: The seasonal Mann-Kendall test is most appropriate because we are observing ozone concentrations over time and that includes over seasonal fluctuations.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a geom_point and a geom_line layer. Edit your axis labels accordingly.

```r
# 13
ggplot(GaringerOzone.monthly, aes(x = Date, y = Ozone.Concentration)) +
  geom_point() +
  geom_line() +
  labs(y = "Monthly Average Ozone Concentration (ppm)",
       title = "Monthly Average Ozone Concentration from 2010 through 2019") +
  geom_smooth(method = 'lm',
              se = TRUE,
              col = '#09BEC8') +
  mytheme_08 +
  theme(plot.title = element_text(size = 08))
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



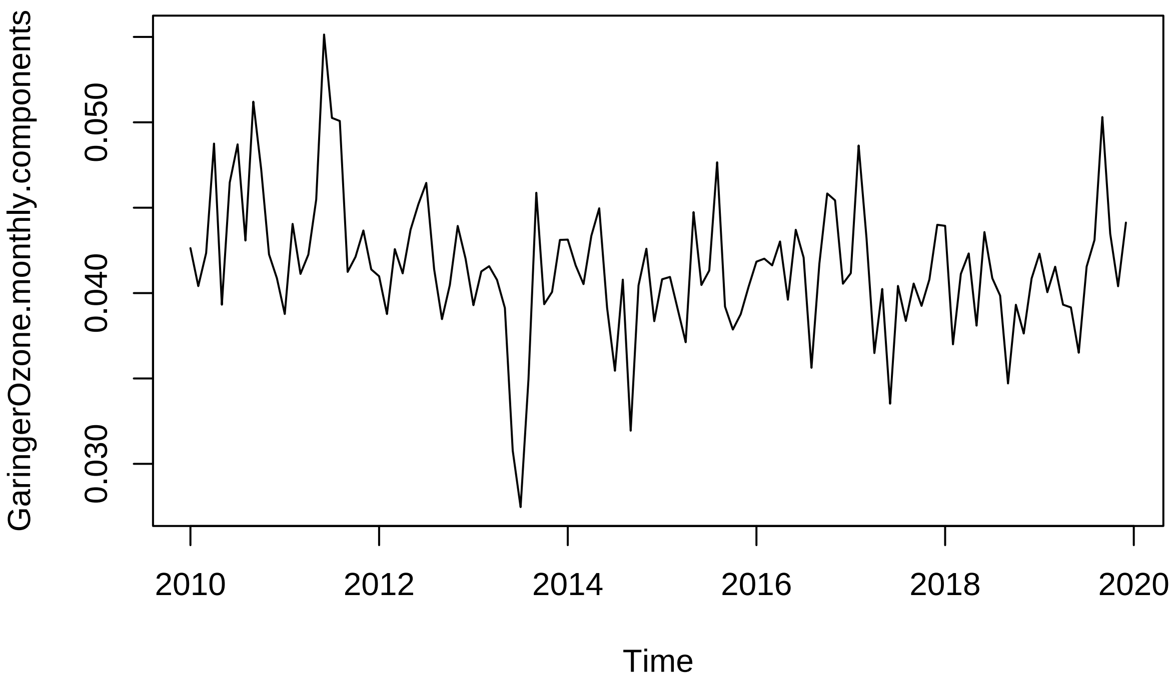
14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

    Answer: Our study question is Have ozone concentrations changed over the 2010s at this station? The plot shows the monthly average ozone concentration from 2010 through 2019. We are studying how the mean ozone concentration changes over time. The seasonal Mann-Kendall trend analysis produced a negatie tau of -0.143 and a p-value of 0.047. The negative tau value tells us that the ozone concentration decreases over time. You can see the gradual change in ozone concentration in our plot. The p-value is less than 0.05 meaning that there is statistical significance to the data. All of which allows us to say yes, ozone concentrations have changed throughout the 2010s.

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the EnoDischarge on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15
GaringerOzone.monthly.components <- (GaringerOzone.monthly.ts - GaringerOzone.monthly.decomposed$time.se

plot(GaringerOzone.monthly.components)
```




```
#16
GaringerOzone.monthly.nonseasonal.trend <- trend::mk.test(GaringerOzone.monthly.components)

GaringerOzone.monthly.nonseasonal.trend
```

```
##
##  Mann-Kendall trend test
##
## data:  GaringerOzone.monthly.components
## z = -2.672, n = 120, p-value = 0.00754
## alternative hypothesis: true S is not equal to 0
```

```
## sample estimates:
##             S          varS           tau
## -1.179000e+03  1.943657e+05 -1.651376e-01
```

Answer: Running the nonseasonal Man-Kendall test gives us a p-value of 0.00754 indicating that there is statistically significance to the variation of data; ozone concentrations have changed over time and is by random chance. The negative tau also shows a downward trend of ozone concentrations. While the nonseasonal test gives different values, both tell us that there is a trend within ozone concentrations that are going in a downward trend.